

FINN 2041 CORPORATE FINANCE

2025/2026

**Dividend and payout
policy**



Learning objectives

- ❑ By the end of the session, you should be able to:
 - explain the ir(relevance) of dividend and payout decisions
 - discuss alternative theories of dividend and payout policy
 - explain the role of dividends in addressing agency problems
 - compare and contrast the effects of dividends and share repurchases
 - apply the Lintner model of dividend smoothing
 - evaluate the implications of different dividend and payout policies

Required readings

- Brealey, R., Myers, S., Allen, F. and Edmans, A. (2022) [Principles of Corporate Finance](#), 14th ed., McGraw Hill: New York., Chapter 15

Introduction

What should
an organisation
invest in?

How does an
organisation fund
its investment?

What is the earnings?

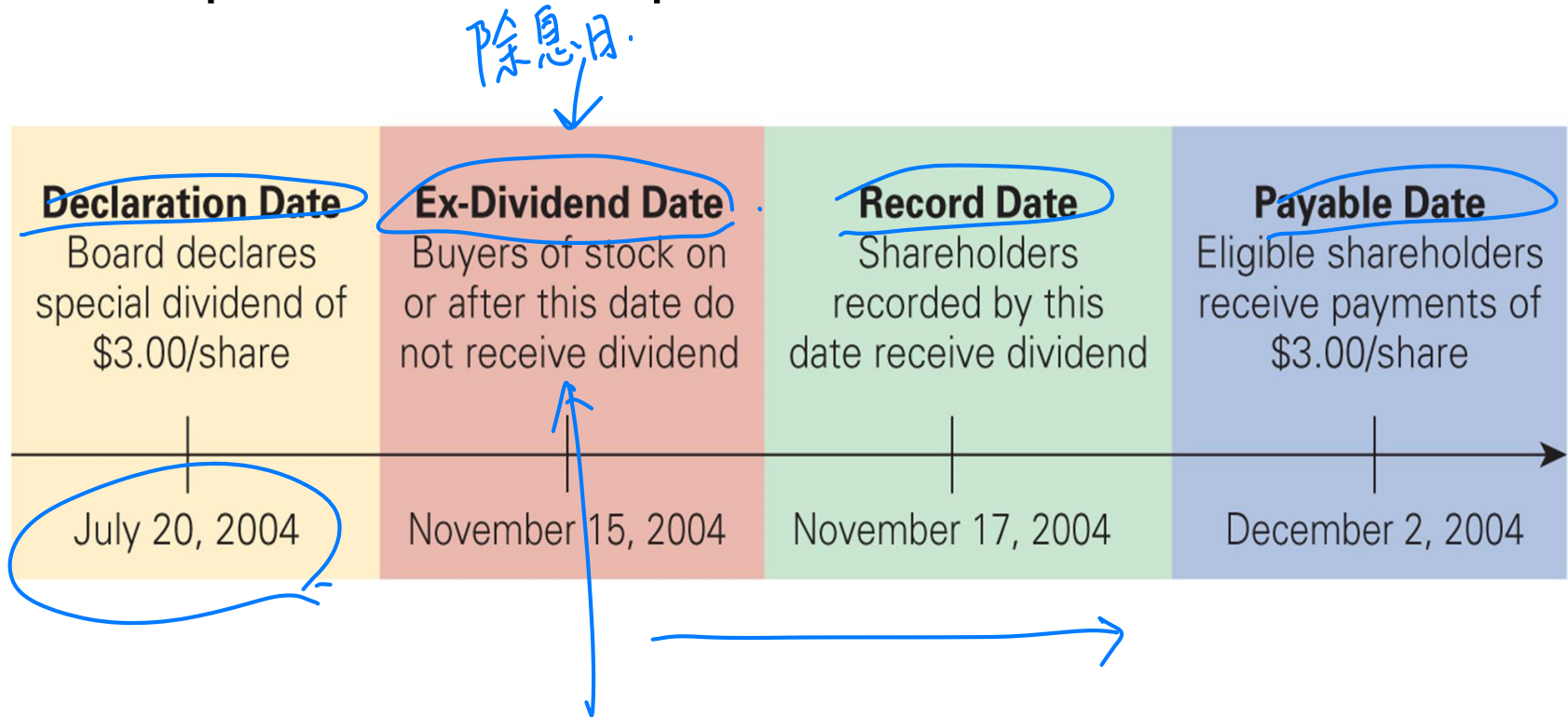
How do we reward the
providers of finance?

Introduction

- ❑ Policy that the firm's management follows in determining the **size** and **pattern** of cash distributions to shareholders over time.
- ❑ The way a firm chooses between the alternative ways to distribute earnings flow to equity holders.
- ❑ A dividend is a payment made to shareholders, mostly at the discretion of the board of directors.

The dividend payment process

□ Example: Microsoft special dividend 2004



Types of dividends

❑ Regular Dividend

- Usually semi-annually, or annually

❑ Special Dividend

- A one-time dividend payment a firm makes, which is usually much larger than a regular dividend

❑ Stock Dividend (Stock Split)

- When a company issues a dividend in shares of stock rather than cash to its shareholders

Miller and Modigliani dividend irrelevance theory

- ❑ In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the firm's value
- ❑ Investors are indifferent between dividends and retention-generated capital gains.

Miller and Modigliani dividend irrelevance theory

- If the firm's cash dividend is too big, you can just take the excess cash received and use it to buy more of the firm's stock. If the cash dividend is too small, you can just sell a little bit of your stock in the firm to get the cash flow you want.

Miller and Modigliani dividend irrelevance theory

□ Homemade dividends

- Shareholders can create a dividend stream to suit themselves

M&M's

**If the company is
not paying a dividend
sell some shares**

**If you don't want
to receive a dividend
buy more shares in
the company**

Homemade dividend example

- $\times \$50 = \$50,000$
- ❑ You own 1,000 shares in a firm that has historically paid dividends at a rate of 50% of earnings per share. Although earnings per share this year are \$5, the firm has decided to retain all of the earnings and not pay a dividend. The current market price is \$50 per share.
- $25 \times 1000 = 2500$

- ❑ How could you create a homemade dividend based on the firm's dividend history?
- $\rightarrow$

MM

Homemade dividend example

Based on the firm's history, its "normal" dividend would have been $\$5 \times 50\% = \2.50 per share.

To create a homemade dividend, you need to sell enough shares to generate the desired cash flow.

Desired cash flow = 1,000 shares $\times$ $\$2.50$ per share = $\$2,500$.

Number of shares to sell at $\$50$ per share = $\$2,500 / \$50 = 50$.

Homemade dividend example

After selling shares to create homemade dividends, the value remaining of the investment = $(1000 \text{ shares} - 50 \text{ shares}) \times \$50 \text{ per share} = \$47,500$ →

Alternatively, if dividends are paid, then the value of the shares is reduced by the amount of dividends = $(\$50 - \$2.5) = \$47.50$

The total value of the investor's portfolio will then be $\$47.50 \times 1000 \text{ shares} = \$47,500$ →

Bird-in-hand theory 一鸟在手理论.

- ❑ Investors think dividends are less risky than potential future capital gains, hence they like dividends.
- ❑ Due to their risk aversion, investors consider current dividends to be less risky
- ❑ If so, investors would value high payout firms more highly.

Dividend signalling

- ❑ Dividend changes reflect managers' views about a firm's future earning prospects (information content).
 - A dividend increase signals to investors that management anticipates sustainable earnings to support the higher payout
 - A dividend decrease may indicate that management is less optimistic about a near-term earnings recovery and aims to conserve cash

Dividend signalling

□ Stock price reaction?

- Generally, dividend increase (initiation) signal good news because it signals managers optimism about the earning prospects.
- Generally, dividend cut (termination) is perceived as bad news because it signals the manager pessimism. ↓

Dividend signalling

□HOWEVER, note that:

- While an increase of a firm's dividend may signal management's optimism regarding its future cash flows, it might also signal a lack of investment opportunities.
- Conversely, a firm might cut its dividend to exploit new positive-NPV investment opportunities. In this case, the dividend decrease might lead to a positive, rather than negative, stock price reaction.

Clientele effect

- ❑ Different groups of investors, or clienteles, prefer different dividend policies.
- ❑ The dividend clientele effect states that high-tax bracket investors (like individuals) prefer low dividend payouts and low tax bracket investors (like corporations and pension funds) prefer high dividend payouts. So different groups desire different levels of dividends.
- ❑ Clientele effects impede changing dividend policy. Taxes & brokerage costs hurt investors who have to switch companies.

Share repurchases

- ❑ Also known as share buyback
- ❑ An alternative way to pay cash to investors is through a share repurchase or buyback.
- ❑ The firm uses cash to buy shares of its own outstanding stock.

Share repurchases

☐ Open Market Repurchase

- When a firm repurchases shares by buying shares in the open market

☐ Tender Offer

- A public announcement of an offer to all existing security holders to buy back a specified amount of outstanding securities at a prespecified price (typically set at a 10% to 20% premium to the current market price) over a prespecified period of time (usually about 20 days)
- If shareholders do not tender enough shares, the firm may cancel the offer, and no buyback occurs.

Share repurchases

☐ Dutch Auction

- Firm lists different prices at which it is prepared to buy shares, and shareholders in turn indicate how many shares they are willing to sell at each price. The firm then pays the lowest price at which it can buy back its desired number of shares.

☐ Targeted Repurchase

- When a firm purchases shares directly from a specific shareholder

☐ Greenmail

Merge

- When a firm avoids a threat of takeover and removal of its management by a major shareholder by buying out the shareholder, often at a large premium over the current market price

Dividend vs share repurchases example

- ❑ Consider Genron Corporation. The firm's board is meeting to decide how to pay out \$20 million in excess cash to shareholders.
- ❑ Genron has no debt, its equity cost of capital equals its unlevered cost of capital of 12%.

Dividend vs share repurchases example

□ With 10 million shares outstanding, Genron will be able to pay a \$2 dividend immediately.

t=1 开始 永远派发 X 元.

permanent

□ The firm expects to generate future free cash flows of \$48 million per year; thus, it anticipates paying a dividend of \$4.80 per share each year thereafter.

• The cum-dividend price of Genron will be

$$P_{cum} = \text{Current Dividend} + PV(\text{Future Dividends}) = 2 + \frac{4.80}{0.12} = 2 + 40 = \$42$$

未来现金流的总和

年金
0 1 2 ... n

t=0.

+ \frac{Coupon}{r.}

Dividend vs share repurchases example

- After the ex-dividend date, new buyers will not receive the current dividend and the share price of Genron will be.

\$2 cash → $\frac{1}{2}$
Δ.

$$\underline{P_{ex}} = PV(\text{Future Dividends}) = \frac{4.80}{0.12} = \$40$$

Dividend vs share repurchases example

	(Cum-Dividend)	(Ex-Dividend)
Cash	20	0
Other assets	400	400
Total market value	420	400
Shares (millions)	10	10
Share price	\$42	\$40

Dividend vs share repurchases example

□ Suppose that instead of paying a dividend this year, Genron uses the \$20 million to repurchase its shares on the open market.

- With an initial share price of \$42, Genron will repurchase 476,000 shares.

- $\$20 \text{ million} \div \$42 \text{ per share} = 0.476 \text{ million shares}$

- This will leave only 9.524 million shares outstanding.

- $10 \text{ million} - 0.476 \text{ million} = 9.524 \text{ million}$

Dividend vs share repurchases example

- Genron's Future Dividends
 - It should not be surprising that the repurchase had no effect on the stock price.
 - After the repurchase, the future dividend would rise to \$5.04 per share.

• \$48 million ÷ 9.524 million shares = \$5.04 per share

- Genron's share price is

$$P_{rep} = \frac{5.04}{0.12} = \$42$$

Dividend vs share repurchases example

	(Before Repurchase)	(After Repurchase)
Cash	20	0
Other assets	400	400
Total market value of assets	420	400
Shares (millions)	10	9.524
Share price	\$42	\$42

Dividend vs share repurchases example

- ❑ In perfect capital markets, an open market share repurchase has no effect on the stock price, and the stock price is the same as the cum-dividend price if a dividend were paid instead.
- ❑ In perfect capital markets, investors are indifferent between the firm distributing funds via dividends or share repurchases. By reinvesting dividends or selling shares, they can replicate either payout method on their own.

Stock dividends and stock splits

□ Stock dividend

- With a stock dividend, a firm does not pay out any cash to shareholders.
- As a result, the total market value of the firm's equity is unchanged. The only thing that is different is the number of shares outstanding.
- The stock price will therefore fall because the same total equity value is now divided over a larger number of shares.

拆股.

1 股拆 2 股
10\$ 1000 股 → 2000 股.
5\$

Stock dividends and stock splits

□ Stock dividend

- Are usually not taxed until the shares are sold.
- Stock dividends are usually expressed as a percentage. Eg 25% stock dividend. This means for every 4 ($100/25$) shares you hold, you get 1 additional share, or for every one share, you get an additional quarter of a share.
- The number of shares is proportionally increased, and the price per share is proportionally reduced so that there is no change in value.

Stock dividends and stock splits

❑ Stock split

- Works similarly to a stock dividend. Company divides the number of existing shares to create more shares.
- Usually expressed as a ratio. E.g. 3 for 1 stock split. This means 1 old share is split into 3 new shares.
Handwritten notes: 2 for 1 (with arrow pointing to 3 for 1), 1股变2股, 100股 -> 300股
- The typical motivation for a stock split is to keep the share price in a range thought to be attractive to small investors.
- If the share price rises “too high,” it might be difficult for small investors to invest in the stock

Stock dividends and stock splits

股票股利 拆股
□ Suppose Genron paid a 50% stock dividend (a 3:2 stock split) rather than a cash dividend.

- A shareholder who owns 100 shares before the dividend has a portfolio worth \$4,200.

$$\textcircled{\$42} \times 100 = \$4,200$$

- After the dividend, the shareholder owns 150 shares. Because the portfolio is still worth \$4,200, the stock price will fall to \$28.

$$\boxed{\$4,200 \div 150 = \$28}$$

Stock dividends and stock splits

- ❑ Cum- and Ex-Dividend Share Price for Genron with a 50% Stock Dividend (\$ million)

	December 11 (Cum-Dividend)	December 12 (Ex-Dividend)
Cash	20	20
Other assets	400	400
Total market value of assets	420	420
Shares (millions)	10	15
Share price	\$42	\$28

Dividend smoothing

- ❑ Companies typically set long-term target ratios of dividends to earnings to maintain stability in the payment of dividends

5\$ 10% 5% 4%

- ❑ A firm is likely to set a low target ratio if it has many positive NPV projects relative to available cash flow and a high ratio if it has few positive NPV projects.
- ❑ Managers need time to assess the permanence of any earnings rise. Hence dividend changes appear to lag earnings changes by a number of periods.

Dividend smoothing

□ The Lintner model

- The stable dividend policy can be represented by a gradual adjustment process

$$Div_1 = Div_0 + [(EPS_1 \times \text{Target payout ratio} - Div_0) \times \text{Adjustment Factor}]$$

Handwritten annotations:
 - Blue arrows point to Div_1 , Div_0 , and EPS_1 .
 - A blue arrow points to the "Adjustment Factor" term.
 - A blue arrow points to the "Target payout ratio" term, with "40%" written above it.
 - A blue arrow points to the EPS_1 term, with "4.5" written above it.
 - A blue circle around the "Adjustment Factor" term, with "4 years" written inside it.
 - A blue circle around the "Adjustment Factor" term, with "1/4" written next to it.

Where:

EPS = Earnings per share

Adjustment factor: number of years required to adjust to dividend amount (eg 5 years = $1/5 = 0.2$)

Dividend smoothing

□ Lintner Model example

Sharpe SA, a company with 2 million shares outstanding, has just paid a dividend of £1.25 per share. The company expects to achieve a dividend payout ratio of 40 percent in four years' time. It has projected that the Earnings Per Share next year will be £4.50. How much cash will the company require to pay dividends next year?

Dividend smoothing

□ Lintner Model solution

$$D_1 = D_0 + [((EPS_1 \times \text{Target payout ratio}) - D_0) \times \text{Adjustment factor}]$$

Four-year adjustment factor = $1/4 = 0.25$

$$1.25 + [((4.50 \times 0.4) - 1.25) \times 0.25]$$

$$= £1.4$$

Cash required = $£1.4 \times 2\text{million shares} = £2.8\text{million}$

Dividends and taxes

- ❑ Recall that personal taxes on equity income take two forms:
 - Personal taxes on dividend income.
 - Personal taxes on capital gains (CG).
- ❑ Shareholders must pay taxes on the dividends they receive and they must also pay CG taxes when they sell their shares.
 - In most countries, dividends are typically taxed at a higher rate than capital gains.
- ❑ Even IF realised CG and dividends incur the same tax rate, investors may prefer a firm NOT to pay a dividend because, the investor can postpone tax on capital gain by delaying the sale of the shares

Dividends and taxes

□ Incorporating personal taxes (T_{PS}) on dividend income into the MM model >> firms should retain all earnings.

- Investor will prefer the cash to be retained and reinvested or used to buyback shares (capital gains from stock appreciation). >> capital gains income will dominate dividend income.
- The irrelevance between payout and retention strategies will disappear in the presence of personal income tax.
x
- The higher tax rate on dividends makes it undesirable for a firm to pay a dividend.

Dividends and taxes

- ❑ Personal income taxes should discourage dividend payments.
- ❑ Firms that pay dividends can have a lower market value.
 - This tax disadvantages will decrease the value of a firm that pay dividends.
- ❑ The optimal dividend policy when the dividend tax rate exceeds the capital gain tax rate is to pay NO dividends at all!

Dividend policy and agency costs

- ❑ Accumulating free cash flow (with significant management discretion) can lead to agency costs
- ❑ When firms have excessive cash, managers may use the funds inefficiently by:
 - Excessive consumption of perquisites, excessive executive perks and compensation, expenses...etc.
 - 'Overinvestment' in negative- or zero- NPV projects, overpaying for acquisitions rather than pay out to shareholders.
 - Reduced employee effort (shirking).
- ❑ Dividends are a way of getting money back to investors, preventing managers from wasting money on negative NPV projects

Dividend policy and agency costs

- ❑ Dividend serve as a bonding and monitoring mechanisms. However, this is not cost free.

- Dividends might lead greater external financing costs
- Borrowing firms face external scrutiny, as they will have to go to the market for additional resources.

- ❑ Another potential conflict that could be affected by dividend policy is between stockholders and bondholders.

- Bondholders would like as much cash as possible to be left in the firm so it can be available during any possible *financial distress*.
- Shareholders may want higher dividends to keep the cash away from bondholders and, in so doing, they may lower the collateral base of the firm.

Types of dividend policies

Strategies adopted by companies to determine how much dividends to pay.

☐ Stable dividend policy

- Paying dividends regularly
- Quite common because managers are reluctant to cut dividends

☐ Constant dividend policy

rate $\times$ earning.

- Based on a fixed payout ratio
- Dividends fluctuate with earnings in the short term

☐ Residual dividend policy

- Dividends paid from remaining internally generated funds after positive NPV projects have been financed.

Types of dividend policies

Residual Dividend Policy Example

Red Zepp Plc follows a strict residual dividend policy. Its debt-equity ratio is 3.

- (a) If earnings for the year are £180,000, what is the maximum amount of capital spending possible with no new equity?
- (b) If planned investment outlays for the coming year are £760,000, will Red Zeppelin pay a dividend? If so, how much?

No.

Types of dividend policies

□ Residual Dividend Policy (Solution)

(a) Since the company has a debt-equity ratio of 3, they can raise £3 in debt for every £1 of equity. The maximum capital outlay with no outside equity financing is:

$$\text{Maximum capital outlay} = £180,000 + 3(£180,000) = £720,000.$$

(b) If planned capital spending is £760,000, then no dividend will be paid.

Why companies continue to pay dividends

❑ Companies continue to pay dividends even though, according to many economists, it should not matter to investors whether a firm pays dividends or not, and notwithstanding the tax disadvantages to investors.

- To signal confidence in the firm's future cash flows
- To cater to the needs of investors with a need for regular income
- To cater to the needs of investors with lower task obligations
- To respond to pressures from shareholders about a stock pile of cash

The end